Dr. Gergana Rusenova

Doctor of Sciences (ETH Zurich)

DIGITAL ENGINEER
ARCHITECTURAL DESIGNER
RESEARCHER AND EDUCATOR

CONTACT

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atireg.github.io/FluxAndFlow

LANGUAGES

Bulgarian · English · German · Russian · Spanish

INTERESTS

Digitalisation · Automation · AI · Sustainable construction · Non-traditional building materials · Pedagogical approaches

SKILLS

Digital engineering
Complex geometry, parametric and algorithmic design, thermal simulation, automation of office and construction processes

Digital fabrication
Robotic fabrication, additive processes, physical prototyping

Web development
JavaScript, React JS, AngularJS, Three.js, MongoDB, responsive web design

AI-assisted development
Building and shipping software with AI coding agents, from the Nap Nap Hooray sleep app to this portfolio website

Research and teaching
Academic writing, curriculum design, subject coordination, PhD supervision

PROFILE

Digital Engineer with over a decade of coding experience, now Associate and Head of Digital Technologies at knippershelbig in Stuttgart. I work on international projects such as airports, yachts and shopping malls, solving complex geometry problems and automating processes, both within the office and for construction.

My path runs from architecture and engineering practice through a doctorate at ETH Zurich in robotic fabrication with low-grade materials, to four years of teaching, research and leadership at Swinburne University of Technology in Melbourne. My research has been published in international journals, conferences and books, and shown at several international exhibition venues.

I enjoy highly interdisciplinary teams, and care for the people I work with.

PROFESSIONAL EXPERIENCE

2023 – current

Associate, Head of Digital Technologies

knippershelbig, Stuttgart

- Developing engineering solutions with code: automating fabrication processes, simulating thermal performance and exploring new material applications in construction.
- Developing strategies for the digitalisation of the office.
- Principal investigator of Frameless Glass Structures, a glass pavilion shown at Glasstec 2024 in Düsseldorf.
- Built kh ReUse, a web platform for cataloguing and reusing building elements salvaged from demolished buildings.

2021 – 2023

Deputy Department Chair for Architecture and Industrial Design

Swinburne University of Technology, Melbourne

- Supported the Department Chair in overseeing six disciplines and over 30 academic staff in the School of Design and Architecture.
- Coordinated marketing events, time-tabling and School meetings; a key role in the accreditation of the Master of Architecture and Urban Design.

2020 – 2021

Director of end-of-year graduation exhibition (GradX) for Architecture and Industrial Design

Swinburne University of Technology, Melbourne

- Reinvented the exhibition online during COVID-19: more than twenty 3D virtual spaces showing the work of nearly 400 graduating students from 18 disciplines.
- 2.2 K visitors from all over the world; Dean's Award for Innovation.

2019 – 2023

Lecturer in Architecture

Swinburne University of Technology, Melbourne

- 50 % teaching, 30 % research, 20 % leadership and service.
- Shaped the newly established bachelor's and master's programs, convening subjects in computational design, robotic fabrication and materiality.
- Co-supervised two PhD students.

2016 – 2019

PhD researcher

Gramazio Kohler Research, ETH Zurich

- Developed computational design and robotic fabrication tools for aggregating textile string and crushed stone into load-bearing, reusable structures with minimal material waste.
- Investigated their load-bearing capacity, and used fibre-optic sensors to measure their structural health.
- Shown at Ars Electronica 2017 and as the Rock Print Pavilion in Winterthur, 2018.

2014 – 2016

Digital Designer

SL Rasch GmbH Special & Lightweight Structures, Stuttgart

- Designed for the model-making department: large-scale, moving and adjustable models for public display of some of the company's largest projects, among them the Makkah Royal Clock Tower.

2013 – 2014

Junior architectural designer

Architectural Office Prof. W. Kergaßner, Stuttgart

- Contributed to the execution phase of the ZF Forum, a 92 million euro, 35,000 m² corporate headquarters.
- Led the design competition for a 250 m speed-testing tower for ThyssenKrupp; second place.

2011 – 2013

Trainee

Knippers Helbig Advanced Engineering, Stuttgart

- Planning and execution for art installations, a museum, an airport, high-rise buildings and bridges in Germany, Finland, Italy, Dubai, Abu Dhabi, the USA and China.

EDUCATION

2023 – 2025

JavaScript Developer Degree

SoftUni, online qualification

- Full-stack web development: JavaScript, React JS and AngularJS, MongoDB, Three.js and responsive web design.
- Final project: kh ReUse. The same knowledge built the portfolio website Flux and Flow.

2016 – 2020

Doctor of Sciences

ETH Zurich, Swiss Federal Institute of Technology

- Thesis: *Material- and Fabrication-informed Design of Structurally-sound Jammed Architectural Structures*.
- Supervised by Prof. Fabio Gramazio and Prof. Matthias Kohler, Chair of Architecture and Digital Fabrication; co-supervised by Dr. Ammar Mirjan and Dr. Romana Rust.

2014 – 2015

Master in Integrative Technologies and Architectural Design Research (ITECH)

University of Stuttgart

- Thesis: *Emergent Space: Simulations and Analyses of Spatial Aggregate Formations*, on the structural behaviour and architectural potential of aggregates, through physical prototypes and material simulations.
- Supervised by Prof. Achim Menges (ICD) and Prof. Jan Knippers (ITKE).

2009 – 2013

Bachelor in Architecture and Urban Planning

University of Stuttgart

- Thesis: *Development of Connections and Structural Reinforcement for Fiber Composites Using Computational Design Methods*, a tool to simulate glass- and carbon-fibre composites and optimise their material for the ICD/ITKE Research Pavilion 2013/14.

TEACHING AND SUPERVISION

2020 – 2022	**Arch Communication 2**, core subject, bachelor · Swinburne, coordinator and tutor
2020 – 2022	**Fabrication and Creative Processes**, core subject, master · Swinburne, coordinator and tutor
2020 – 2022	**earth for Earth!**, design research studio, master · Swinburne, coordinator and tutor
2021	**Up in the Air**, design research studio, master · Swinburne, coordinator and tutor
2020	**Construction 2**, design studio · Swinburne, co-coordinator and tutor
2018	**Master thesis**, Dept. of Civil Engineering · ETH Zurich, co-supervisor
2017	**Master of Advanced Studies in Architecture and Digital Fabrication** · ETH Zurich, tutor

PhD co-supervision, Swinburne: *Programming a material system for resilient architecture* (School of Design and Architecture, from 2022) and *Development of reusable mould system for the fabrication of concrete shells* (School of Engineering, from 2021).

SELECTED PROJECTS AND EXHIBITIONS

2024	**Frameless Glass Structures**, Glasstec, Düsseldorf · principal investigator
2022	**Hortiscape: a robotically crafted living wall**, CAADRIA, Sydney · research and project lead
2022	**Future Gardens**, student exhibition, Melbourne · convenor and tutor
2018	**Rock Print Pavilion**, "Hello, Robot", Winterthur · research and project lead
2017	**Rock Print: A Manistone**, Ars Electronica Festival, Linz · research and project lead
2015	**ICD Aggregate Pavilion 2015**, Stuttgart · research assistant
2014	**ICD/ITKE Research Pavilion 2013/14**, Stuttgart · "System and Development" team

GRANTS, AWARDS AND TALKS

2023	**Invited talk**, Shapeshift Festival, a festival for technology, science and contemporary culture, Sofia
2022	**Grant, AUD 42,199**, flat-pack DIY garden sheds and backyard offices for AKD Softwood Pty Ltd · research lead and grant holder
2022	**Adobe Innovation Grant, AUD 5,000**, accessible robotics through digital and physical prototyping · research lead and grant holder
2022	**Conference panel chair**, CAADRIA 2022, Session 7B Robotics and Digital Fabrication in Construction
2021	**UA-DAAD grant, AUD 10,000**, textile-reinforced granular materials in architecture (Australia-Germany Joint Research Cooperation) · research lead and grant holder

2021	**Mention in *The Economist***, “Knitting a road with stones and string”: Empa built on the doctoral research for road building
2021	**Invited talk**, Melbourne University, ADD+F Research Hub
2021	**Workshop host**, robotic fabrication with COMPAS FAB, with Gramazio Kohler Research, ETH Zurich
2020	**Dean’s Award for Innovation**, reinvention of GradX 2020, Swinburne
2020	**Invited talk**, “Fragile Matter in Architecture”, Swinburne Design Lecture Series
2017	**Ars Electronica STARTS Prize**, Grand Prize – Innovative Collaboration
2017	**Invited talk**, “Young Successful Researchers”, ETH Zurich, one of three PhD researchers selected

SELECTED PUBLICATIONS

1. Rusenova, G., Länge, J., Fildhuth, T., Buksak, A., Peters, T., Ayvaz, I., Haller, M., Schieber, R., Knaack, U. (2026). Integrated Thermal and Fabrication Workflow for Structural Frameless Glass Structures. *Challenging Glass Conference Proceedings*, 10, TU Delft. Article 678
2. Ayvaz, I., Peters, T., Fildhuth, T., Rusenova, G., Buksak, A., Haller, M., Schuster, M., Knaack, U., Kraus, M. (2025). Structural performance of linearly laminated metal fittings for frameless glass shell structures. *Glass Structures & Engineering*, 10, 27. doi:10.1007/s40940-025-00312-4
3. Surti, M., Sanjayan, J., Rusenova, G. (2025). Effectiveness of chain-mail mat for improving actuators efficiency in the pin-and-bed adaptive moulding. *Construction and Building Materials*, 460. doi:10.1016/j.conbuildmat.2024.139755
4. Rusenova, G., Sadar, J., Bohnenberger-Fehr, S., Chen, C. (2023). Unearthing sustainable material futures. In *Sustainability and Toxicity of Building Materials*, Elsevier.
5. Melles, F., Paay, J., Woodcock, I., Rusenova, G. (2021). Practical understandings and use of smart city concepts in Australia. *ANZAScA 2020*, 54th International Conference of the Architectural Science Association, pp. 875–884.
6. Aejmelaeus-Lindström, P., Rusenova, G., Mirjan, A., Medina Ibanez, J., Gramazio, F., Kohler, M. (2020). Rock Print Pavilion: Robotically fabricating architecture from rock and string. *Construction Robotics*.
7. Rusenova, G., Aejmelaeus-Lindström, P., Mayer, H., Gramazio, F., Kohler, M. (2020). Jammed Architectural Structures. In *Data, Matter, Design: Strategies in Computational Design*, Taylor & Francis / Routledge.
8. Rusenova, G. (2019). *Material- and Fabrication-informed Design of Structurally-sound Jammed Architectural Structures*. Doctoral thesis, ETH Zurich.
9. Aejmelaeus-Lindström, P., Rusenova, G., Mirjan, A., Gramazio, F., Kohler, M. (2019). Direct Deposition of Jammed Architectural Structures. *ROBARCH 2018*, Zurich, pp. 270–281.
10. Rusenova, G., Wittel, F. K., Aejmelaeus-Lindström, P., Gramazio, F., Kohler, M. (2018). Load-Bearing Capacity and Deformation of Jammed Architectural Structures. *3D Printing and Additive Manufacturing*, 5(4), pp. 257–267.
11. Rusenova, G., Dierichs, K., Baharlou, E., Menges, A. (2016). Feedback- and Data-driven Design for Aggregate Architectures. *ACADIA 2016: Posthuman Frontiers*, Ann Arbor, pp. 62–72.

References available on request.